



Trinity Grammar School

Mathematics Department

2015

TRIAL EXAMINATION

HSC ASSESSMENT TASK 4

Year 12

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- BOSTES approved calculators for this course may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations
- Write your BOSTES Student Number on this question paper and on any answer sheets or writing booklets used to write your responses
- If you do not attempt a question you must submit an answer sheet or writing booklet for that question clearly indicating N/A and your BOSTES Student Number
- HSC Assessment Weighting: 40%

BOSTES Student Number

--	--	--	--	--	--	--	--

Class Teacher:

.....

Do NOT write solutions to be marked on this question paper. Any working on this question paper will NOT be marked.

Date of Examination:

Thursday, 13 August 2015

Total marks – 100

Section I

10 marks

- Attempt Questions 1–10
- Allow about **15** minutes for this section

Section II

90 marks

- Attempt Questions 11–16
- Allow about **2 hours 45** minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Shade the correct response on the multiple-choice answer sheet for Questions 1–10

Each question is worth 1 mark

1 If $z = 1 + 2i$ and $w = i - 3$ then $z - \bar{w}$ is?

(A) $3i - 2$

(B) $4 + 3i$

(C) $i - 2$

(D) $4 + i$

2 The function defined by $f(x) = \sqrt{3} \cos x + 3 \sin x$ has an amplitude of

(A) $3 - \sqrt{3}$

(B) $\sqrt{3}$

(C) $2\sqrt{3}$

(D) $3 + \sqrt{3}$

3 The equations of the directrices of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are

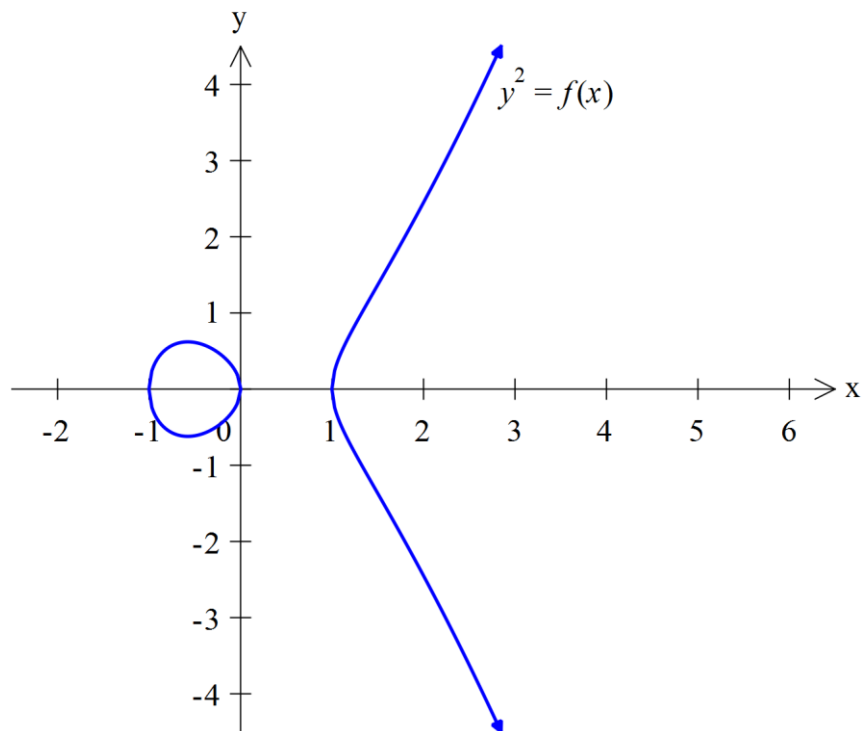
(A) $x = \pm \frac{16\sqrt{7}}{7}$

(B) $x = \pm \frac{\sqrt{7}}{16}$

(C) $x = \pm \frac{16}{5}$

(D) $x = \pm \frac{5}{16}$

- 4 The diagram below shows the graph of $y^2 = f(x)$.



Which expression best represents the function $f(x)$?

- (A) $x^2(x-1)$
- (B) $x^2(1+x)$
- (C) $x(x^2-1)$
- (D) $x(1+x^2)$
- 5 The linear factorisation of $x^4 + 7x^2 - 18$ over the complex number field is:

- (A) $(x-i\sqrt{2})(x+i\sqrt{2})(x-3)(x+3)$
- (B) $(x-i\sqrt{2})(x-i\sqrt{2})(x+3)(x+3)$
- (C) $(x-3i)(x-3i)(x+\sqrt{2})(x+\sqrt{2})$
- (D) $(x-3i)(x+3i)(x-\sqrt{2})(x+\sqrt{2})$

6 If α, β, γ are the roots of $x^3 + x - 1 = 0$, then the cubic equation with roots $\frac{\alpha}{2}, \frac{\beta}{2}, \frac{\gamma}{2}$ is

(A) $-8x^3 - 2x - 1 = 0$

(B) $8x^3 + 2x - 1 = 0$

(C) $\frac{8}{x^3} + \frac{2}{x} - 1 = 0$

(D) $\frac{x^3}{8} + \frac{x}{2} - 1 = 0$

7 The circle $x^2 + y^2 = 4$ is rotated about the line $x = 3$ through 360° . The volume of the solid formed is given by

(A) $V = 2\pi \int_{-2}^2 (3-x)\sqrt{4-x^2} \, dx$

(B) $V = 4\pi \int_{-2}^2 (3-x)\sqrt{4-x^2} \, dx$

(C) $V = 2\pi \int_0^2 (3-x)\sqrt{4-x^2} \, dx$

(D) $V = 4\pi \int_0^2 (4-x)\sqrt{3-x^2} \, dx$

8 The equation $x^2 - y^2 - 2xy - 2 = 0$ defines y implicitly as a function of x .

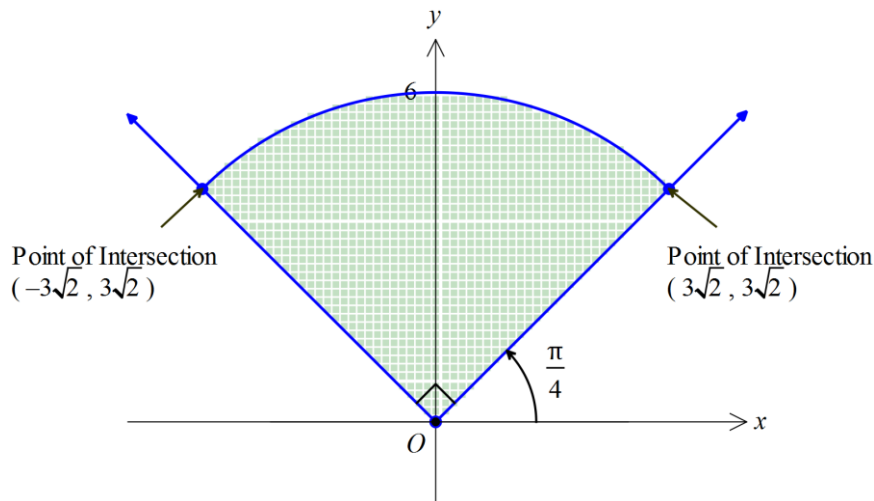
The value of $\frac{dy}{dx}$ at the point $(-\sqrt{2}, 0)$ is?

(A) $-\sqrt{2}$

(B) -1

(C) 0

(D) 1



Which pair of inequalities represents the shaded region of the Argand plane shown above?

- (A) $|z| \leq 6$ and $\frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$
- (B) $|z| \geq 6$ and $\frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$
- (C) $|z| \geq 36$ and $-\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{4}$
- (D) $|z| \geq 36$ and $\frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$

10 What is the natural domain of the function $f(x) = \frac{1}{2} \left(x\sqrt{x^2 - 1} - \log_e \left(x + \sqrt{x^2 - 1} \right) \right)$?

- (A) $x \leq -1$
- (B) $-1 \leq x \leq 1$
- (C) $x \leq -1$ or $x \geq 1$
- (D) $x \geq 1$

End of Section I
Section II commences on the next page

Section II

90 marks

Attempt Questions 11–16

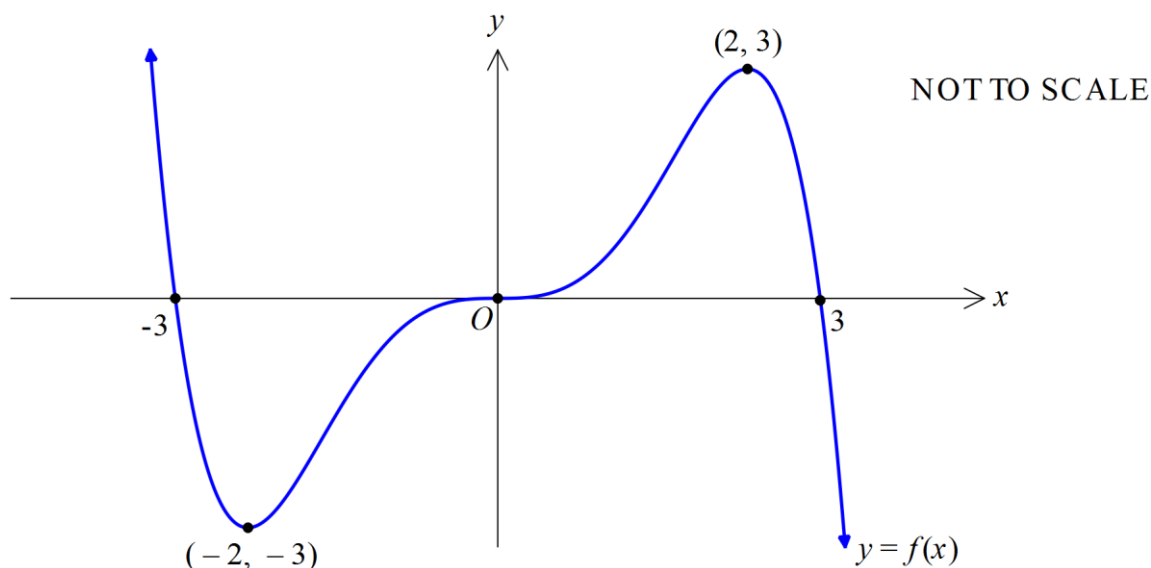
Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include all relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) Consider the curve $y = f(x)$, with its stationary points and intercepts shown in the diagram below.



- (i) **On separate one-third page number planes**, sketch the graphs of:

(α) $y = \frac{1}{f(x)}$ 2

(β) $y = \sqrt{f(x)}$ 2

(γ) $|y| = \sqrt{f(x)}$ 2

(δ) $y = e^{f(x)}$ 2

- (ii) State the number of solutions to the equation $\sqrt{f(x)} = 1$. 1

Question 11 continues on the following page...

Question 11 continued

(b) Sketch the locus of the complex number z such that:

(i) $|z - 2i| \leq |z - 2|$ 2

(ii) $0 \leq \arg(z - 1 - i) \leq \frac{\pi}{6}$ 2

(iii) $\operatorname{Im}\left(z - \frac{1}{z}\right) = 0, z \neq 0.$ 2

End of Question 11

Please turn over

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) (i) Show that $2+i$ is a root of $x^3 - 11x + 20 = 0$. 2
- (ii) Find all the roots of $x^3 - 11x + 20 = 0$. 2
- (b) Given that α, β and γ are the roots of $x^3 + 2x^2 - x - 3 = 0$, find the value of $\alpha^2 + \beta^2 + \gamma^2$. 2
- (c) Given the polynomial $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$ has a root of multiplicity 3, find all the roots of the equation $P(x) = 0$. 3
- (d) (i) Write down all the fifth roots of -1 in the form $\cos \theta + i \sin \theta$ ($-\pi < \theta \leq \pi$), **and** plot the five roots on an Argand diagram. 3
- (ii) Hence, express $x^5 + 1$ as the product of real factors in the form $(x+1)(x^2 + (a_1 \cos b_1 \pi)x + c_1)(x^2 + (a_2 \cos b_2 \pi)x + c_2)$ where a_1, a_2, c_1, c_2 are integers and b_1, b_2 are rational numbers. 3

End of Question 12

Please turn over

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1+x^2}} dx$ using the substitution $x = \tan \theta$. **4**

(b) Find:

(i) $\int_{-1}^1 \frac{3}{5-2x+x^2} dx$ **3**

(ii) $\int x e^{-2x} dx$ **2**

(c) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on the rectangular hyperbola $xy = c^2$. Tangents to the rectangular hyperbola at P and Q intersect at the point R .

(i) Prove that the coordinates of R in terms of c , p and q are $\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$. **3**

(ii) If P and Q are variable points on the rectangular hyperbola which move such that $p^2 + q^2 = 2$, show that the equation of the locus of R is $y^2 = 2c^2 - xy$, stating any restriction(s) that may apply. **3**

End of Question 13

Please turn over

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) A fly is standing on a small turntable at a distance of 8 cm from the centre. If the turntable is rotating at 45 revolutions per minute, find
- (i) the speed of the fly in metres per second (correct to two decimal places); 2
 - (ii) the acceleration of the fly (correct to two decimal places). 1
- (b) The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the ellipse meets the x -axis at the points $A(a, 0)$ and $A'(-a, 0)$.
- (i) Prove that the equation of the tangent at P is given by $bx \cos \theta + ay \sin \theta = ab$. 2
 - (ii) The tangent at P meets the tangents from A and A' at points Q and Q' respectively. Find the coordinates of Q and Q' . 2
 - (iii) The points A, A', Q' and Q form a trapezium. Prove that the product of the lengths of the parallel sides is independent of the position of P . 2
- (c) (i) Write down the expansion of $(\cos \theta + i \sin \theta)^3$ in the form $a + ib$, where a and b are in terms of $\sin \theta$ and $\cos \theta$. 2
- (ii) Hence show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. 2
- (iii) **Hence** solve the equation $\cos 3\theta + \cos \theta = 0$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. 2

End of Question 14

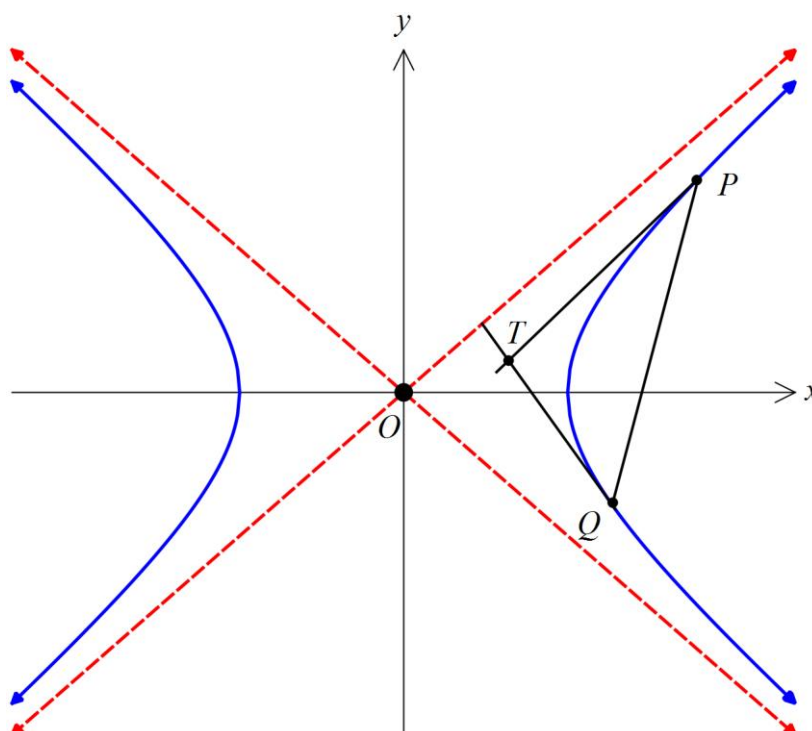
Please turn over

Question 15 (15 marks) Use a SEPARATE writing booklet

(a) Suppose ω is one of the complex roots of $z^3 - 1 = 0$.

(i) Show that $1 + \omega + \omega^2 = 0$. 1

(ii) Show that $\left(\frac{\omega}{1+\omega}\right)^k + \left(\frac{\omega^2}{1+\omega^2}\right)^k = (-1)^k 2 \cos \frac{2k\pi}{3}$, where k is a positive integer. 3



(b) In the diagram above, the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on the same branch of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The tangents at P and Q meet at $T(x_0, y_0)$.

(i) Write down, in Cartesian form, the equation of the tangent at P . 1

(ii) Hence show that the chord of contact, PQ , has equation $\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$. 2

(iii) The chord PQ passes through the right-hand focus $S(ae, 0)$, where e is the eccentricity of the hyperbola. 2

Prove that T lies on the directrix of the hyperbola.

Question 15 continues on the next page...

Question 15 continued

(c) (i) Find $\int_0^{\frac{\pi}{2}} \cos x \, dx$. **1**

(ii) Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$. **3**

Show, using integration by parts, $I_n = \frac{n-1}{n} I_{n-2}$ (n is an integer ≥ 2).

(iii) Hence, evaluate $I_5 = \int_0^{\frac{\pi}{2}} \cos^5 x \, dx$. **2**

End of Question 15

Please turn over

Question 16 (15 marks) Use a SEPARATE writing booklet

- (a) (i) Find real numbers p , q and r such that 3

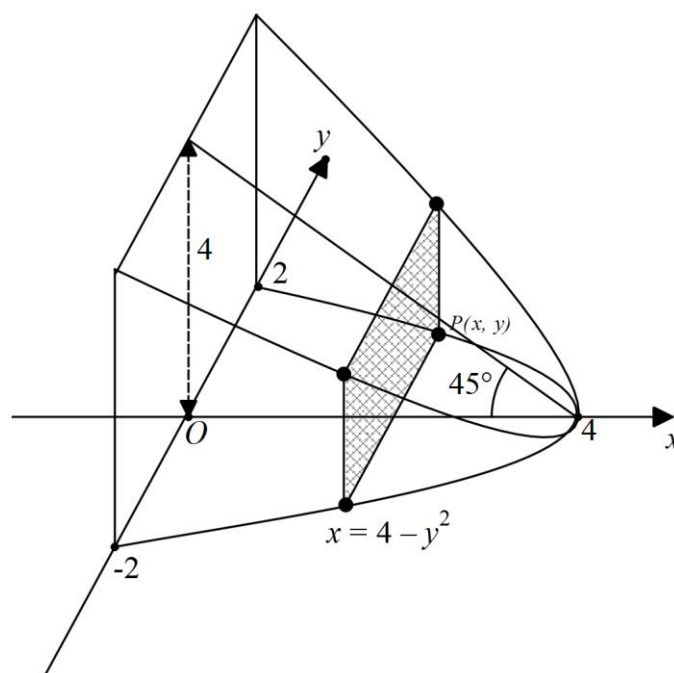
$$\frac{300x}{x^3 + 1000} = \frac{p}{x + 10} + \frac{qx + r}{x^2 - 10x + 100}$$

- (ii) A particle of mass m kg is projected vertically upwards in a highly resistive medium at an initial velocity of 5 m/s. The particle is subjected to the force of gravity and to a resistance due to the medium of magnitude $\frac{1}{100}mv^3$.
Given the acceleration due to gravity is 10 m/s², x is the displacement of the particle from its starting position, and regarding upwards motion as positive

- (α) Explain why $\ddot{x} = -\frac{1}{100}(1000 + v^3)$. 1

- (β) Hence, find the maximum height reached by the particle, giving your answer correct to one decimal place. 4

- (b) The base of a solid is the region enclosed by the parabola $x = 4 - y^2$ and the y -axis.

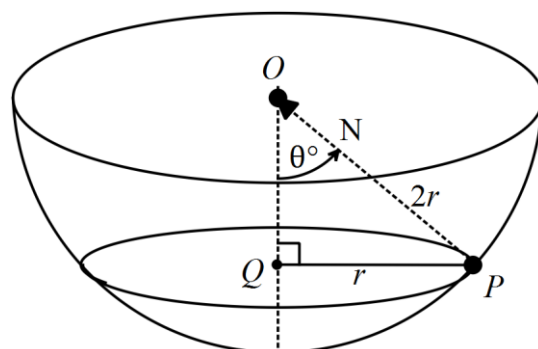


The top of the solid is formed by a plane inclined at 45° to the x - y plane. Each vertical cross-section of the solid taken parallel to the y -axis is a rectangle as shown in the diagram above. The point $P(x, y)$ lies on $x = 4 - y^2$. 4

Find the volume of the solid.

Question 16 is continued on the next page

Question 16 continued



- (c) The diagram above shows a particle P , of mass m kg moving with constant angular velocity ω in a horizontal circle about Q , of radius r metres on the inside of a fixed smooth hemispherical bowl of internal radius $2r$ metres, centre at O . There is a normal reaction force N acting inward toward the centre of the hemisphere.

3

By resolving forces vertically and radially at P , show that $\omega^2 = \frac{g}{r\sqrt{3}}$.

End of Question 16

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

Blank page



Trinity Grammar School
Mathematics Department

2015

Year 12 Mathematics Extension 2

HSC Assessment Task 4

Section I

Objective Response Answer Sheet

BOSTES Student Number

--	--	--	--	--	--	--	--	--

Class Teacher: _____

EH

- Write your answers for **Section I** on this answer sheet.
- **Shade completely only ONE answer for each question.**
- To change an answer, erase your previous mark completely and then record your new answer.

1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)
5	(A)	(B)	(C)	(D)
6	(A)	(B)	(C)	(D)
7	(A)	(B)	(C)	(D)
8	(A)	(B)	(C)	(D)
9	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)

A table of Standard Integrals is printed on the reverse side of this answer sheet

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Shade the correct response on the multiple-choice answer sheet for Questions 1–10

Each question is worth 1 mark

- 1 If $z = 1 + 2i$ and $w = i - 3$ then $z - \bar{w}$ is?

$$\bar{w} = -3 - i$$

$$\therefore z - \bar{w} = 1 + 2i - (-3 - i) \\ = 4 + 3i$$

(A) $3i - 2$

~~(B) $4 + 3i$~~

(C) $i - 2$

(D) $4 + i$

- 2 The function defined by $f(x) = \sqrt{3} \cos x + 3 \sin x$ has an amplitude of

(A) $3 - \sqrt{3}$

(B) $\sqrt{3}$

~~(C) $2\sqrt{3}$~~

(D) $3 + \sqrt{3}$

$$f_{\max} = \sqrt{(\sqrt{3})^2 + 3^2} = 2\sqrt{3}$$

$$f_{\min} = -\sqrt{(\sqrt{3})^2 + 3^2} = -2\sqrt{3}$$

$$\therefore \text{Amp} = \frac{2\sqrt{3} + (-2\sqrt{3})}{2} \\ = 2\sqrt{3}$$

- 3 The equations of the directrices of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are

~~(A) $x = \pm \frac{16\sqrt{7}}{7}$~~

(B) $x = \pm \frac{\sqrt{7}}{16}$

(C) $x = \pm \frac{16}{5}$

(D) $x = \pm \frac{5}{16}$

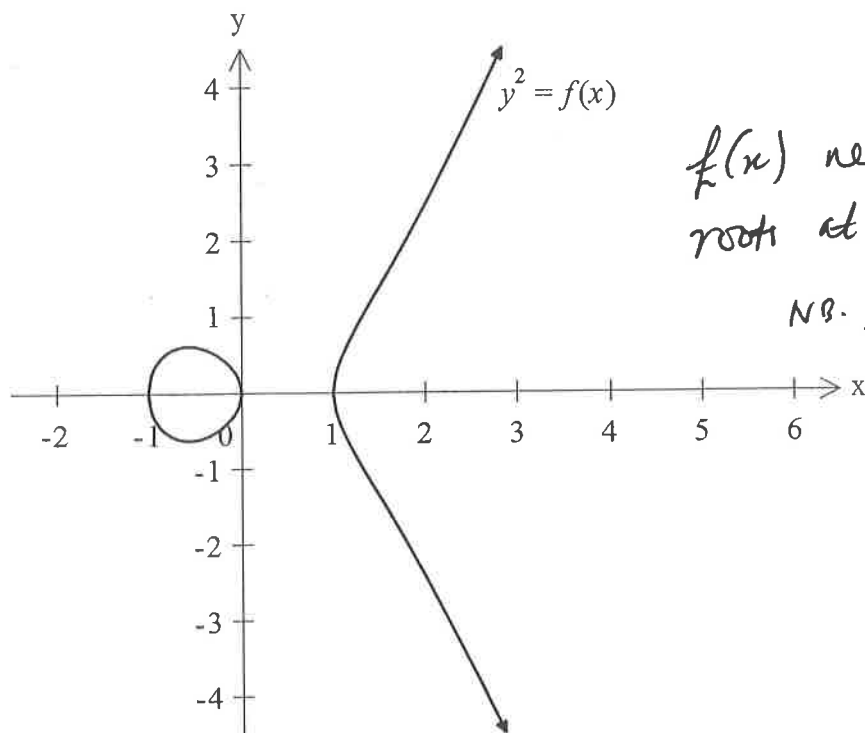
$$a = 4, b = 3 \\ b^2 = a^2(1 - e^2)$$

$$9 = 16(1 - e^2)$$

$$\therefore e^2 = \frac{7}{16}$$

$$\therefore x = \pm \frac{a}{e} = \pm \frac{4}{\sqrt{7/4}} \\ = \pm \frac{16}{\sqrt{7}}$$

- 4 The diagram below shows the graph of $y^2 = f(x)$.



$f(x)$ needs to have roots at $-1, 0, 1$.

NB. $\lim_{x \rightarrow \infty} y \rightarrow \pm \infty$.

Which expression best represents the function $f(x)$?

(A) $x^2(x-1)$

(B) $x^2(1+x)$

~~(C)~~ $x(x^2-1) \Rightarrow x=0, x=\pm 1$.

(D) $x(1+x^2)$

- 5 The linear factorisation of $x^4 + 7x^2 - 18$ over the complex number field is:

(A) $(x-i\sqrt{2})(x+i\sqrt{2})(x-3)(x+3)$

(B) $(x-i\sqrt{2})(x-i\sqrt{2})(x+3)(x+3)$

(C) $(x-3i)(x-3i)(x+\sqrt{2})(x+\sqrt{2})$

~~(D)~~ $(x-3i)(x+3i)(x-\sqrt{2})(x+\sqrt{2})$

$$\begin{aligned} &= (x^2+9)(x^2-2) \\ &= (x^2-3i)^2(x^2-2) \\ &= (x-3i)(x+3i)(x-\sqrt{2})(x+\sqrt{2}) \end{aligned}$$

- 6 If α, β, γ are the roots of $x^3 + x - 1 = 0$, then the cubic equation with roots $\frac{\alpha}{2}, \frac{\beta}{2}, \frac{\gamma}{2}$ is

(A) $-8x^3 - 2x - 1 = 0$

~~(B)~~ $8x^3 + 2x - 1 = 0$

(C) $\frac{8}{x^3} + \frac{2}{x} - 1 = 0$

(D) $\frac{x^3}{8} + \frac{x}{2} - 1 = 0$

let $y = \frac{x}{2}$ i.e. $x = 2y$
subst: $(2y)^3 + 2y - 1 = 0$

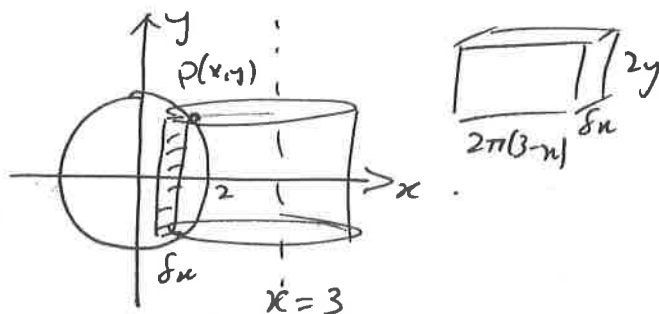
- 7 The circle $x^2 + y^2 = 4$ is rotated about the line $x = 3$ through 360° . The volume of the solid formed is given by

(A) $V = 2\pi \int_{-2}^2 (3-x)\sqrt{4-x^2} dx$

~~(B)~~ $V = 4\pi \int_{-2}^2 (3-x)\sqrt{4-x^2} dx$

(C) $V = 2\pi \int_0^2 (3-x)\sqrt{4-x^2} dx$

(D) $V = 4\pi \int_0^2 (4-x)\sqrt{3-x^2} dx$



$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-2}^2 2\pi(3-x)2y \delta x$$

$$= 4\pi \int_{-2}^2 (3-x)\sqrt{4-x^2} dx$$

- 8 The equation $x^2 - y^2 - 2xy - 2 = 0$ defines y implicitly as a function of x .

The value of $\frac{dy}{dx}$ at the point $(-\sqrt{2}, 0)$ is?

(A) $-\sqrt{2}$

(B) -1

(C) 0

~~(D)~~ 1

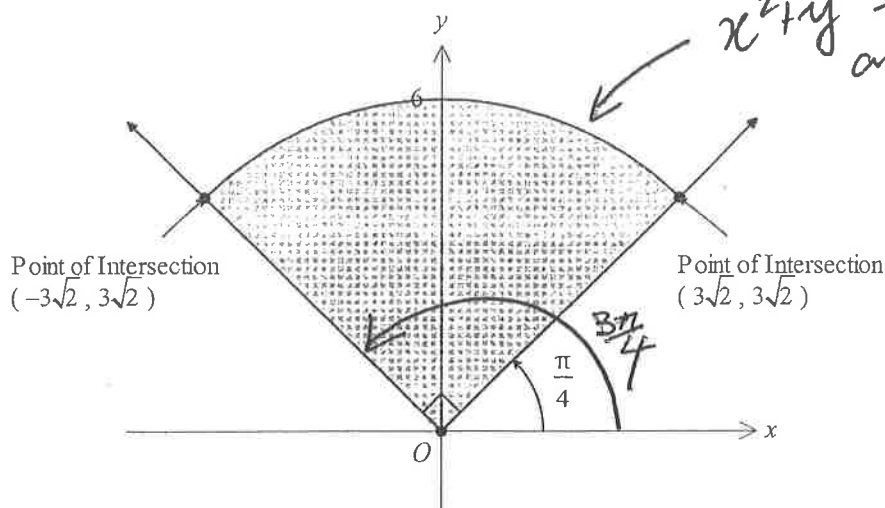
$$2x - 2y \frac{dy}{dx} - 2 \left[y + x \frac{dy}{dx} \right] = 0$$

at $(-\sqrt{2}, 0)$

$$-2\sqrt{2} - 2 \left[0 - \sqrt{2} \frac{dy}{dx} \right] = 0$$

$$\therefore \frac{dy}{dx} = \frac{2\sqrt{2}}{2\sqrt{2}}$$

$$= 1$$



Which pair of inequalities represents the shaded region of the Argand plane shown above?

- ~~(A)~~ $|z| \leq 6$ and $\frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$
- (B) $|z| \geq 6$ and $\frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$
- (C) $|z| \geq 36$ and $-\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{4}$
- (D) $|z| \geq 36$ and $\frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$

10 What is the natural domain of the function $f(x) = \frac{1}{2} \left(x\sqrt{x^2-1} - \log_e \left(x + \sqrt{x^2-1} \right) \right)$?

- (A) $x \leq -1$
- (B) $-1 \leq x \leq 1$
- (C) $x \leq -1$ or $x \geq 1$

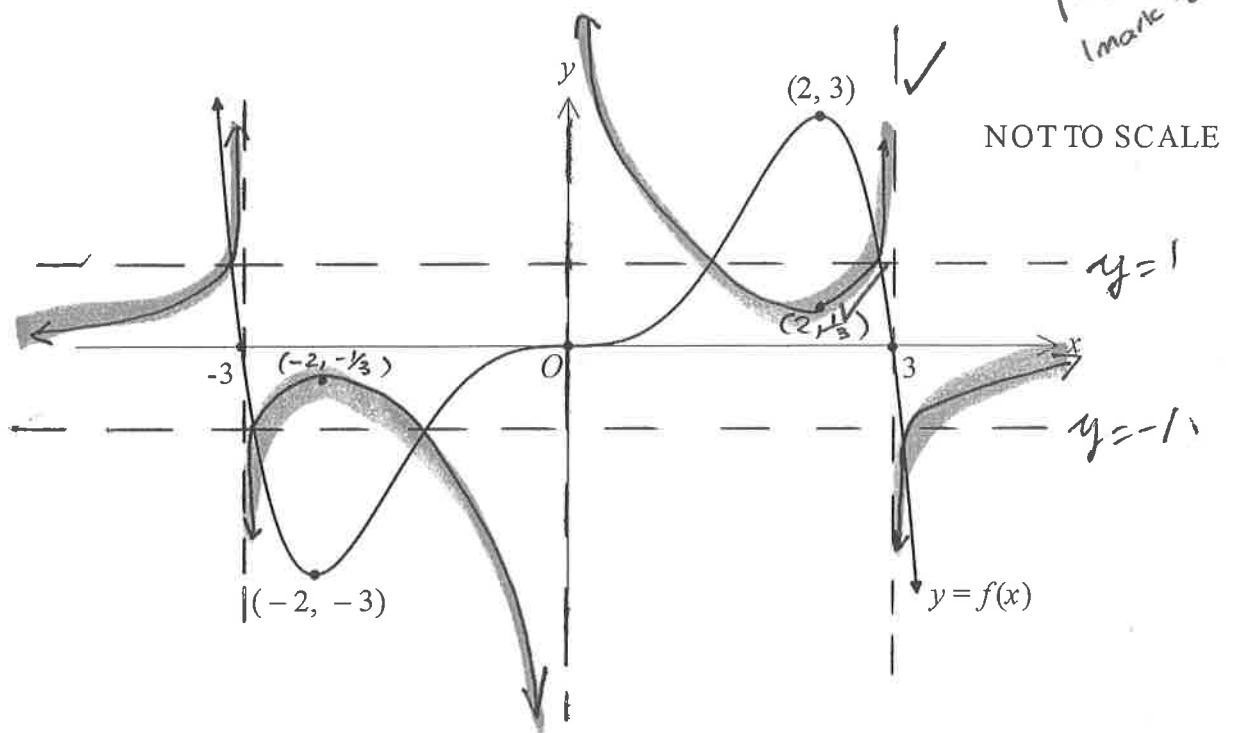
~~(D)~~ $x \geq 1$

want $x^2 - 1 \geq 0$ and $x \geq 0$
 $\downarrow$
 $x \leq -1$ or $x \geq 1$ and $x \geq 0$
 $\therefore x \geq 1$ only

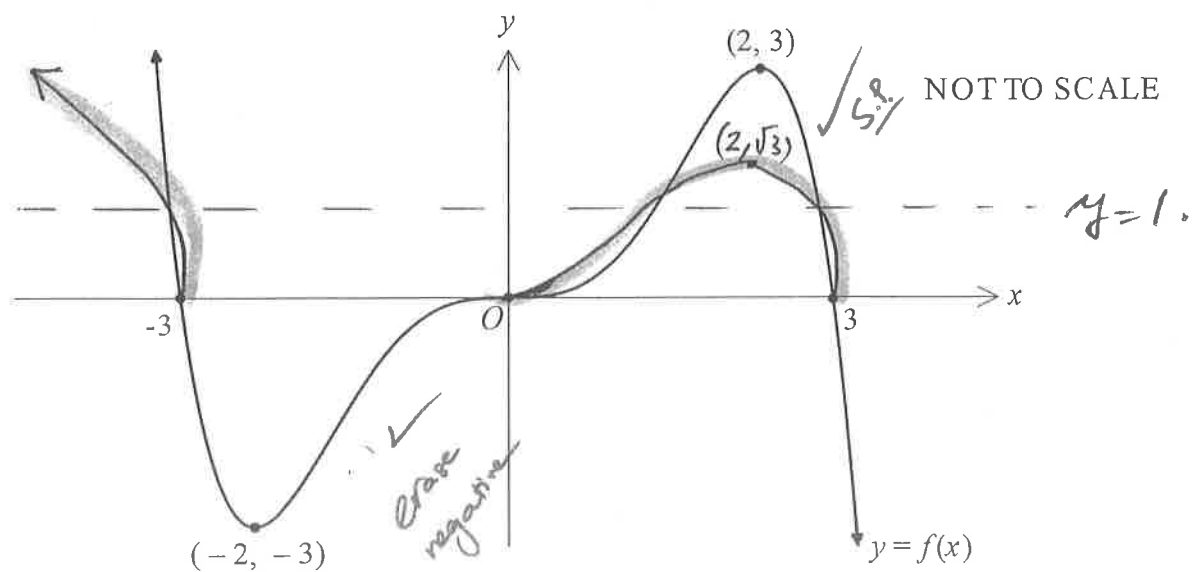
End of Section I
 Section II commences on the next page

2015 HSC MAXX Task 4 Question 11 (a)

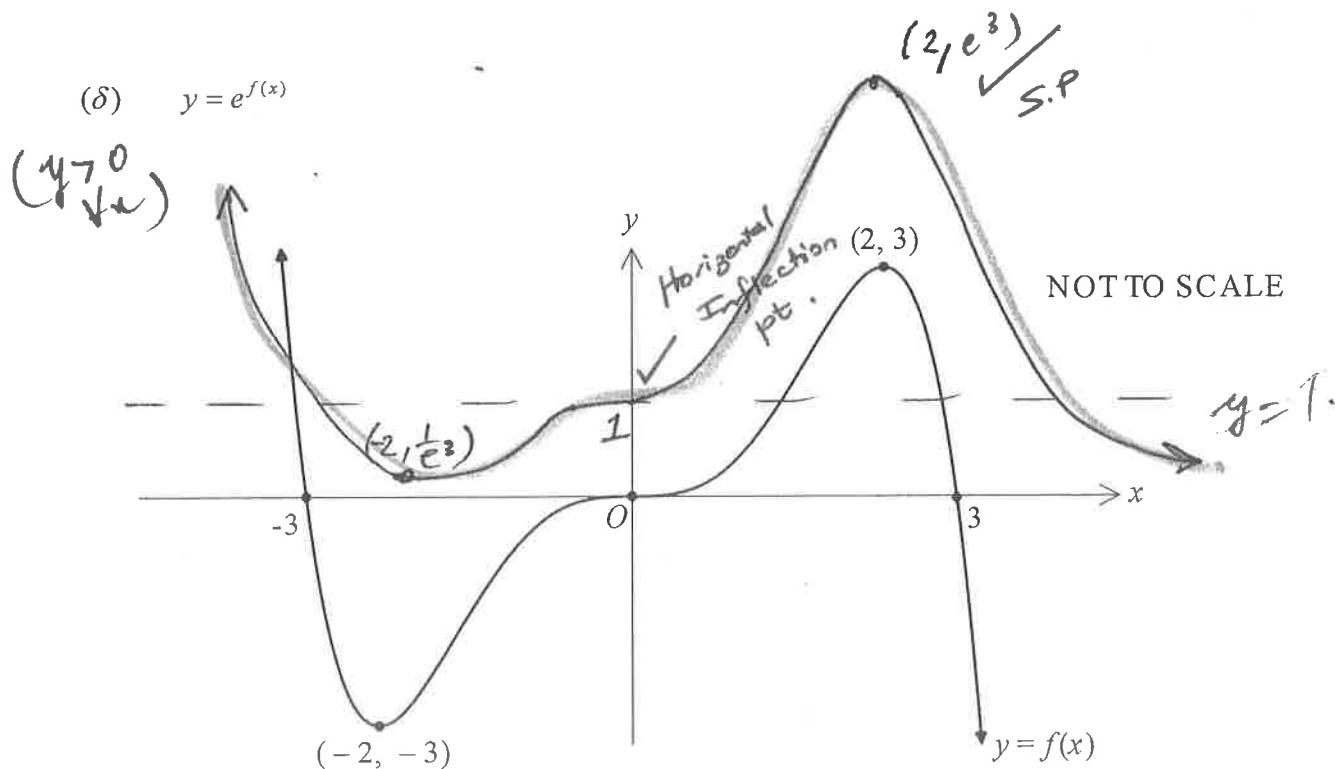
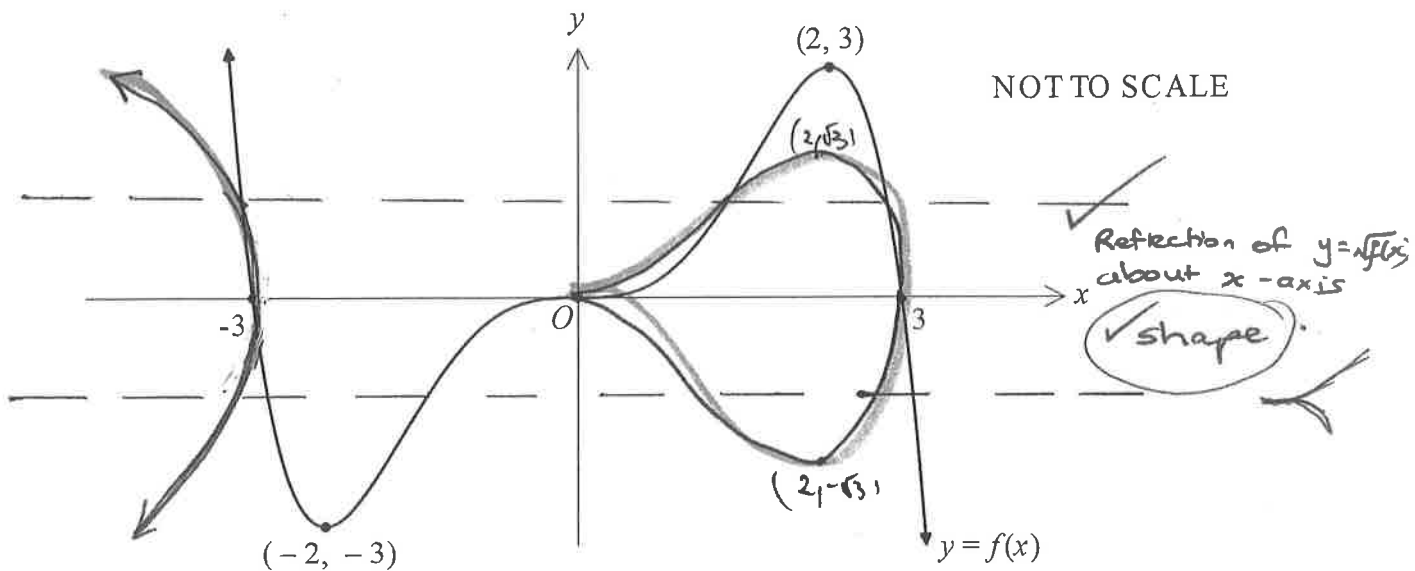
(i) (a) $y = \frac{1}{f(x)}$



(b) $y = \sqrt{f(x)}$

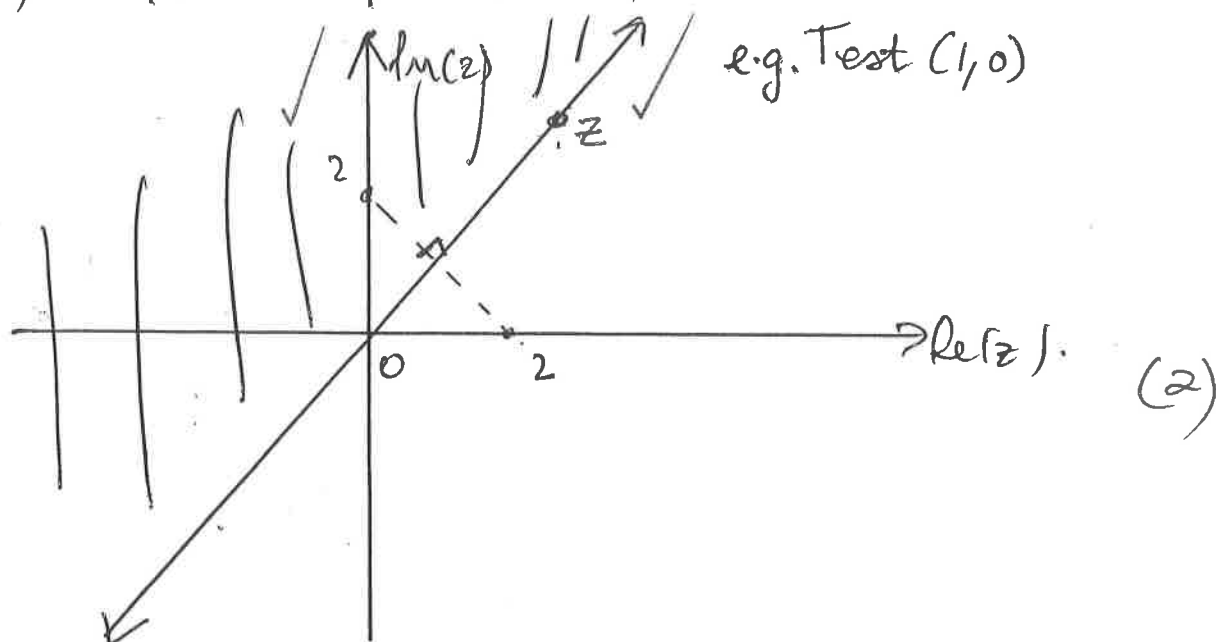


(γ) $|y| = \sqrt{f(x)}$

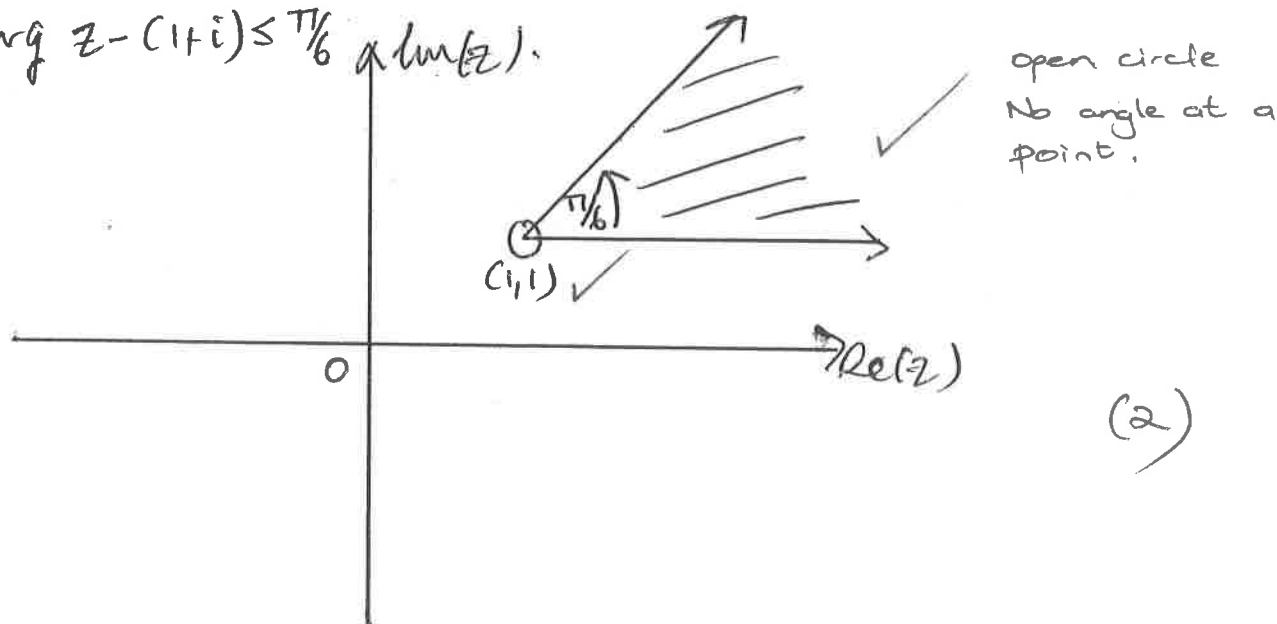


iii) 3 first draw $y=1$ or $y = \sqrt{f(x)}$.

11b) i) $|z - 2i| \leq |z - 2|$



ii) $\arg z - (1+i) \leq \pi/6$



iii) $\ln(z - \frac{1}{z}) = 0, z \neq 0$

$\Rightarrow \arg(z - \frac{1}{z}) = n\pi$

$n \in \mathbb{Z}$

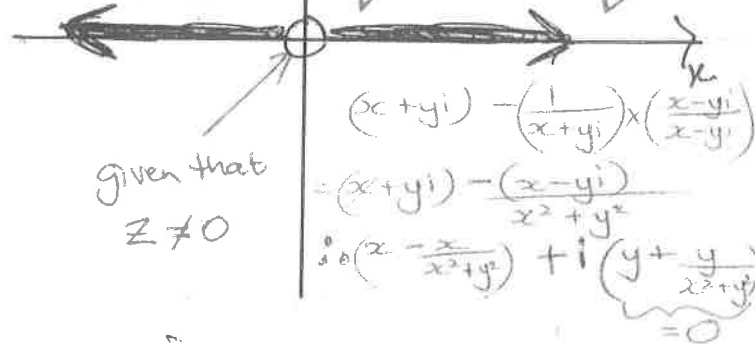
must lie on real axis ($z \neq 0$).

NB: $(x+yi) - (\frac{1}{x+yi})$
 $= (x - \frac{x}{x^2+y^2}) + i(\frac{y}{x^2+y^2} + y)$

want $\frac{y}{x^2+y^2} + y = 0$

$\Rightarrow y = 0$ or $x^2 + y^2 = -1$ impossible in $\mathbb{R}$

$$\begin{aligned} & \left(\frac{\cos \theta + i \sin \theta}{\cos \theta + i \sin \theta} - \frac{1}{\cos \theta + i \sin \theta} \right) \times \frac{\cos \theta - i \sin \theta}{\cos \theta - i \sin \theta} \\ &= \frac{(\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta)}{(\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta)} \\ &= \frac{2i \sin \theta}{\cos^2 \theta + \sin^2 \theta} = \frac{2i \sin \theta}{1} = 2i \sin \theta = 0 \\ &\therefore 2 \sin \theta = 0 \\ &\theta = 0 + n\pi \end{aligned}$$



12) a) i) let $p(x) = x^3 - 11x + 20$.

$$\begin{aligned} p(2+i) &= (2+i)^3 - 11(2+i) + 20 \\ &= 8 + 12i - 6 - i - 22 - 11i + 20 \\ &= 0 + 0i \\ &= 0 \end{aligned}$$

(2)

Since $p(2+i) = 0$ then $x = 2+i$ is a root of $p(x) = 0$.

ii) The coefficients of $p(x)$ are all real and since $x = 2+i$ is a root then $x = 2-i$ (conjugate) is also a root.

Using product of roots

$$(2+i)(2-i)\alpha = -20$$

$$5\alpha = -20$$

$$\therefore \alpha = -4$$

$\therefore$ All the roots of $p(x) = 0$ are $2+i$, $2-i$ and -4 .

[There are alternative methods].

b) The cubic with roots $\alpha^2, \beta^2, \gamma^2$

$$\text{is } (\sqrt{x})^3 - 2(\sqrt{x})^2 - \sqrt{x} - 3 = 0$$

$$\underline{\text{ce}} \quad x\sqrt{x} - \sqrt{x} = 2x + 3$$

$$\underline{\text{ce}} \quad x(x-1) = (2x+3)^2$$

$$\underline{\text{ce}} \quad x^3 - 6x^2 - 11x - 9 = 0$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = 6 \quad (-b/a)$$

Alternatively:

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= (-2)^2 - 2(-1) \\ &= 6 \end{aligned}$$

OR
Sum of Roots
 $2+i+2-i+\alpha = 0$
 $4+\alpha = 0$
 $\alpha = -4$

Note:
OR $(x-(2+i))(x-(2-i))$
 $= x^2 - 2x + ix - 2ix + 4 + 1$
 $= x^2 - 4x + 5$

Long division:
 $\begin{array}{r} x+4 \overline{) x^2-4x+5} \\ \underline{x^2+4x+16} \\ -8x-11 \\ \underline{-8x-32} \\ 21 \end{array}$
 $x^2 - 4x + 5 \overline{) x^3 + 0x^2 - 11x + 20}$
 $\underline{-x^3 + 4x^2 + 5x}$
 $4x^2 - 16x + 20$
 $\underline{4x^2 - 16x + 20}$
 0

(2)

To find $\alpha^2 + \beta^2 + \gamma^2$

$$\text{let } y = x^2 \\ \text{ie } x = \sqrt{y}$$

$$p(\sqrt{y}) = y\sqrt{y} + 2y - \sqrt{y} - 3 = 0$$

$$(\sqrt{y} - \sqrt{y})^2 = (2y+3)^2$$

$$y^3 - 2y^2 + y = 4y^2 - 12y + 9$$

$$y^3 - 6y^2 + 13y - 9 = 0$$

Now for $\alpha^2 + \beta^2 + \gamma^2$

$$\text{find } \left(-\frac{b}{a}\right) = \frac{-(-6)}{1} = 6$$

(2)

c) Given $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$

$$P'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$P''(x) = 24x^2 + 54x + 12$$

Solving $24x^2 + 54x + 12 = 0$

ie $4x^2 + 9x + 2 = 0$

$$(4x+1)(x+2) = 0$$

$$\Rightarrow x = -\frac{1}{4}, x = -2 \quad \checkmark$$

testing $P'(-2) = 8(-2)^3 + 27(-2)^2 + 12(-2) - 20$
 $= -64 + 108 - 24 - 20$
 $= 0$

$$P(-2) = 2(-2)^4 + 9(-2)^3 + 6(-2)^2 - 20(-2) - 24$$

$$= 0$$

This must be the triple root.

By the product of roots

$$(-2)(-2)(-2)(\alpha) = -\frac{24}{2}$$

$$-8\alpha = -\frac{24}{2}$$

$$\alpha = \frac{3}{2} \quad \checkmark \quad (3)$$

$\therefore$ All the roots are $-2, -2, -2$, and $\frac{3}{2}$.

d) i) The fifth roots of -1 are equally spaced on the circle $x^2 + y^2 = 1$ by $\frac{2\pi}{5}$ about 0. One of the ^{5th} roots of (-1) is $-1 \cdot \sqrt[5]{1} = -1$. Others can be given by $x = \cos\left(\frac{k\pi}{5}\right)$
 $k = 0, \pm 1, \pm 2$ (general form)

(or) $x = \cos \pi, \cos \frac{3\pi}{5}, \cos(-\frac{3\pi}{5}), \cos \frac{\pi}{5}, \cos(-\frac{\pi}{5})$
 $(= -1)$ ↑ ↑ ↑ ↑
conjugates conjugates

d) $z^5 = -1$
 $(\cos \theta + i \sin \theta)^5 = -1$
 $\cos 5\theta + i \sin 5\theta = -1$
 $\therefore \cos 5\theta = -1$
 $5\theta = \pi + 2k\pi$
 $\theta = \frac{\pi + 2k\pi}{5}$
 let $k = 0, 1, 2, -1, -2$
 $z_1 = \cos \frac{\pi}{5} \quad (k=0)$
 $z_2 = \cos \frac{3\pi}{5} \quad (k=1)$
 $z_3 = \cos \pi \quad (k=2)$
 $z_4 = \cos(-\frac{\pi}{5}) \quad (k=-1)$
 $z_5 = \cos(-\frac{3\pi}{5}) \quad (k=-2)$

⑬ a) let $x = \tan \theta$ when $x=1, \theta = \pi/4$
 $\frac{dx}{d\theta} = \sec^2 \theta$ $x = \sqrt{3}, \theta = \pi/3$
 $x^2 = \tan^2 \theta$ } ✓

$$\therefore \int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1+x^2}} \frac{dx}{d\theta} d\theta$$

$$= \int_{\pi/4}^{\pi/3} \frac{\sec^2 \theta d\theta}{\tan^2 \theta \sqrt{\sec^2 \theta}} = \frac{\sec \theta}{\tan^2 \theta}$$

$$= \int_{\pi/4}^{\pi/3} \sec \theta \cot^2 \theta d\theta = \int_{\pi/4}^{\pi/3} \frac{1}{\cos \theta} \times \left(\frac{\cos^2 \theta}{\sin^2 \theta} \right) d\theta$$

$$= \int_{\pi/4}^{\pi/3} \cot \theta \csc \theta d\theta = \int_{\pi/4}^{\pi/3} \left(\frac{\cos \theta}{\sin \theta} \right) \times \left(\frac{1}{\sin \theta} \right) d\theta$$

$$= -\csc \theta \Big|_{\pi/4}^{\pi/3}$$

$$= -\csc \pi/3 + \csc \pi/4$$

$$= \sqrt{2} - \frac{2}{\sqrt{3}}$$

NB: $\frac{d}{dx} \sec x = \sec x \tan x$

$\therefore \frac{d}{dx} \csc x = \frac{d}{dx} \sec(\pi/2 - x)$

$$= \sec(\pi/2 - x) \cdot (-1) = -\csc x \cot x$$

b) i) $3 \int_1^1 \frac{1}{4 + (x-1)^2} dx$ ✓

$$= \frac{3}{2} \tan^{-1} \left(\frac{x-1}{2} \right) \Big|_1^1$$

$$= \frac{3}{2} [\tan^{-1} 0 - \tan^{-1} (-1)]$$

$$= \frac{3}{2} \times \pi/4$$

$$= 3\pi/8$$

I.B.P

$$u = x \quad v' = e^{-2x}$$

$$u' = 1 \quad v = \frac{1}{-2} e^{-2x}$$

$$x \left(-\frac{1}{2} e^{-2x} \right) + \int \frac{1}{2} e^{-2x} dx$$

$$= -\frac{x}{2} e^{-2x} - \frac{1}{4} e^{-2x} + C$$

(3)

ii) $\int x e^{-2x} dx = -\frac{x}{2} e^{-2x} - \frac{1}{4} e^{-2x} + C$ (2)
 $= -\frac{1}{2} e^{-2x} \left(x + \frac{1}{2} \right) + C$

13(c)

i) Tangent at P:

$$x + p^2 y = 2cp \quad (1)$$

Tangent at Q:

$$x + q^2 y = 2cq \quad (2)$$

$$(1) - (2) \quad (p^2 - q^2)y = 2cp - 2cq$$

$$\Rightarrow y = \frac{2c(p-q)}{(p-q)(p+q)} \quad (p \neq q)$$

$$y = \frac{2c}{p+q} \quad (3)$$

Sub (3) into (1)

$$x = 2cp - p^2 \left(\frac{2c}{p+q} \right)$$

$$x = \frac{2cp(p+q) - 2cp^2}{p+q}$$

$$x = \frac{2cpq}{p+q} \quad (4)$$

$$\therefore R = \left[\frac{2cpq}{p+q}, \frac{2c}{p+q} \right] \quad (3)$$

ii) from (3) and (4)

$$y^2 = \frac{4c^2}{(p+q)^2} \quad (5)$$

$$= \frac{4c^2}{p^2 + q^2 + 2pq}$$

$$= \frac{4c^2}{2 + 2pq}$$

$$y^2 = \frac{4c^2}{2(1+pq)}$$

$$\text{now in (4) } pq = \frac{x(p+q)}{2c}$$

$$\text{but } (p+q) = \frac{2c}{y} \text{ from (3)}$$

$$pq = x/y \quad \text{both}$$

$$\text{hence } y^2 = \frac{4c^2}{2(1+pq)}$$

$$y^2 = \frac{4c^2}{2(1+x/y)}$$

$$y^2 = \frac{2c^2 y}{x+y}$$

$$y^2(x+y) = 2c^2$$

$$y^2 = \frac{2c^2 - xy}{x+y} \quad (\text{as required})$$

Possible Restrictions:

$$\frac{y}{x+y} > 0; x \neq -y; p \neq -q$$

$$pq \neq -1; y \neq 0$$

OR

$$x+y$$

$$= \frac{2cpq}{p+q} + \frac{2c}{p+q}$$

$$xy = \frac{4c^2 pq}{(p+q)^2} \quad (1)$$

$$y^2 = \frac{4c^2}{(p+q)^2} \quad (2)$$

$$xy + y^2 = \frac{4c^2 pq + 4c^2}{(p+q)^2} \quad (1) + (2)$$

$$= \frac{4c^2(pq+1)}{(p+q)^2} = \frac{4c^2(pq+1)}{2(1+pq)} = 2c^2$$

Given 3 marks

14) a) i) $45 \text{ rpm} = \frac{45 \times 2\pi}{60} \text{ rad/s}$

$\therefore \omega = \frac{3\pi}{2} \text{ rad/s}$ ✓

using $v = r\omega$

$= 0.08 \times \frac{3\pi}{2}$ ✓

$= 0.38 \text{ m/s}$
(2dp) ✓

ii) using $a = r\omega^2$
 $= 0.08 \times \left(\frac{3\pi}{2}\right)^2$
 $= 1.78 \text{ m/s}^2$ (2dp) ✓

b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{b^2}{a^2} \left(\frac{x}{y}\right)$

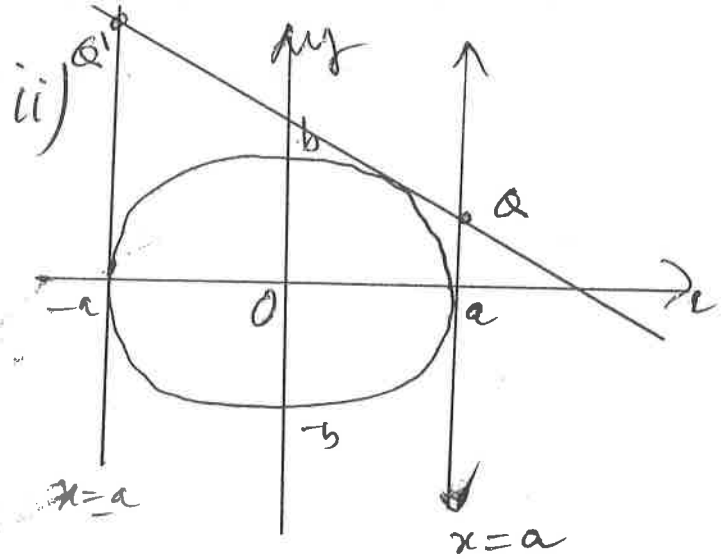
at P, $\frac{dy}{dx} = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta}$
 $= -\frac{b}{a} \frac{\cos \theta}{\sin \theta}$ ✓

using $y - y_1 = m(x - x_1)$

$y - b \sin \theta = -\frac{b}{a} \frac{\cos \theta}{\sin \theta} (x - a \cos \theta)$

$\Rightarrow ay \sin \theta - ab \sin^2 \theta$
 $= -b \cos \theta x + ab \cos^2 \theta$ ✓

$\therefore bx \cos \theta + ay \sin \theta = ab$



at $x = \pm a$

$\pm ab \cos \theta + ay \sin \theta = ab$

$ay \sin \theta = ab \mp ab \cos \theta$

$y = \frac{b \pm b \cos \theta}{\sin \theta} \quad (a \neq 0)$

$\therefore Q = \left(a, \frac{b + b \cos \theta}{\sin \theta}\right)$ ✓

$Q' = \left(-a, \frac{b - b \cos \theta}{\sin \theta}\right)$ ✓

iii) $AQ \cdot Q'A'$

$= \left| \frac{b(1 + \cos \theta)}{\sin \theta} \right| \left| \frac{b(1 - \cos \theta)}{\sin \theta} \right|$ ✓

$= \frac{b^2 (1 - \cos^2 \theta)}{\sin^2 \theta}$

$= \frac{b^2 \sin^2 \theta}{\sin^2 \theta}$

$= b^2$ ✓

$\therefore$ since $|AQ| \cdot |Q'A'|$ is independent of θ then the product is independent of position P.

c) i)

$$\begin{aligned}(e^{i\theta})^3 &= c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 \\&= \cos^3\theta - 3\cos^2\theta\sin\theta i - 3\cos\theta\sin^2\theta - i\sin^3\theta \\&= (\cos^3\theta - 3\cos\theta\sin^2\theta) + i(3\cos^2\theta\sin\theta - \sin^3\theta)\end{aligned}$$

ii) by de Moivre's Theorem

$$(\cos\theta)^3 = \cos 3\theta$$

$$\begin{aligned}\therefore \cos 3\theta &= \operatorname{Re} [(e^{i\theta})^3] \checkmark \\&= \cos^3\theta - 3\cos\theta\sin^2\theta \\&= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta) \\&= 4\cos^3\theta - 3\cos\theta\end{aligned}$$

iii) $\cos 3\theta + \cos\theta = 0$

$$4\cos^3\theta - 3\cos\theta + \cos\theta = 0$$

$$\cos\theta [4\cos^2\theta - 2] = 0$$

$$\cos\theta = 0 \text{ or } \cos\theta = \pm \frac{1}{\sqrt{2}}$$

$$\theta = \pm \frac{\pi}{2} \checkmark \quad \theta = \pm \pi/4 \checkmark$$

$$[-\pi/2 \leq \theta \leq \pi/2].$$

(15)

a) $z^3 - 1 = 0$

i) $(z-1)(z^2+z+1) = 0$ ✓

$z=1$ or $z^2+z+1=0$

if ω is a root $[\omega = \cos(\pm \frac{2\pi}{3})]$

$\therefore \omega^2 + \omega + 1 = 0$

ii) $LHS = \left(\frac{\omega}{1+\omega}\right)^k + \left(\frac{\omega^2}{1+\omega^2}\right)^k$

$= \left(\frac{\omega}{-\omega^2}\right)^k + \left(\frac{\omega^2}{-\omega}\right)^k$ ✓ from (i)

$= (-1)^k \omega^{-k} + (-1)^k \omega^k$ ✓

$= (-1)^k (\omega^{-k} + \omega^k)$

but $z^n + z^{-n} = 2 \cos n\theta$

$\therefore LHS = (-1)^k \left[2 \cos \frac{2\pi k}{3} \right]$

since $\omega = \cos \frac{2\pi}{3}$ in this requirement.

b) i) $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$ ✓

ii) tangent at Q is

$\frac{xx_2}{a^2} - \frac{yy_1}{b^2} = 1$ ✓

but T (x_0, y_0) lies on both TP and TQ

therefore its coordinates

Satisfy both tangents at P and Q.

Let $\frac{x_0 x_1}{a^2} - \frac{y_0 y_1}{b^2} = 1 \rightarrow (P)$

$\frac{x_0 x_2}{a^2} - \frac{y_0 y_2}{b^2} = 1 \rightarrow (Q)$

replacing (x, y) in either

(P) & (Q)

$\Rightarrow \frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$ ✓

iii) Sub $x = ae, y = 0$

$\frac{x_0 ae}{a^2} = 1$ ✓

$\therefore x_0 = \frac{a}{e}$

which is the equation of the directrix.

c) i) $\int_0^{\pi/2} \cos x \, dx$

$= \sin x \Big|_0^{\pi/2}$

$= \sin \frac{\pi}{2} - \sin 0$

$= 1$ ✓

$$\text{ii)} \quad I_n = \int_0^{\pi/2} \cos^n x \, dx = \int_0^{\pi/2} \cos x \cdot \cos^{n-1} x$$

$$u = \cos^{n-1} x$$

$$dv = \cos x$$

$$du = -(n-1) \cos^{n-2} x (-\sin x) \quad v = \sin x$$

$$I_n = \sin x \cos^{n-1} x \Big|_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^2 x \cos^{n-2} x \, dx$$

$$= (n-1) \int_0^{\pi/2} (1 - \cos^2 x) \cos^{n-2} x \, dx \quad \checkmark$$

$$I_n = (n-1) [I_{n-2} - I_n]$$

$$I_n + (n-1)I_n = (n-1)I_{n-2} \quad \checkmark$$

$$I_n = \left(\frac{n-1}{n}\right) I_{n-2} \quad n \geq 2$$

$$\text{iii)} \quad I_5 = \int_0^{\pi/2} \cos^5 x \, dx$$

$$= \frac{4}{5} I_3 \quad \checkmark$$

$$= \frac{4}{5} \times \frac{2}{3} \times I_1$$

$$= \frac{4}{5} \times \frac{2}{3} \times 1$$

$$= \frac{8}{15}$$

only if (i) is correct.

$I_5 > 0$ as well
irrespective

from (i)

16

ai) Want

$$300x \equiv p(x^2 - 10x + 100) + (x+10)(9x+r)$$

Sub $x = -10$

$$-3000 = p(100 + 100 + 100)$$

$$p = -10 \quad \checkmark$$

Sub $x = 0$

$$0 \equiv 100p + 10r$$

$$\therefore 0 = -1000 + 10r$$

$$10r = 1000$$

$$r = 100 \quad \checkmark$$

Sub $x = 10$

$$3000 = p(100 - 100 + 100) + 20(100 + r)$$

$$3000 = 100p + 20(100 + 100)$$

$$3000 = -1000 + 2000 + 200r$$

$$\frac{4000 - 2000}{200} = r$$

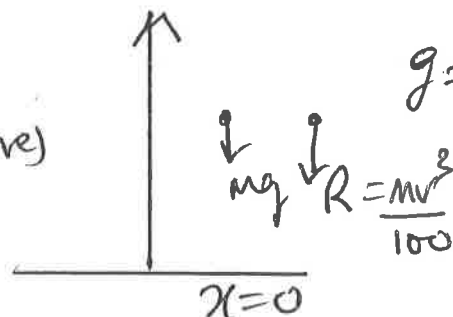
$$r = \frac{2000}{200}$$

$$= 10 \quad \checkmark$$

$$\therefore (p, r) = (-10, 10, 100)$$

ii)

(+ve)



$$g = 10 \text{ m/s}^2$$

(x)

$$t = 0, v = 5 \text{ m/s}$$

by Newton's 2nd Law

$$m\ddot{x} = \sum F_x$$

$$m\ddot{x} = -mg - \frac{mv^2}{100} \quad \checkmark$$

$$\ddot{x} = -g - \frac{v^2}{100}$$

$$\ddot{x} = -\frac{1}{100}(100 \times 10 + v^2)$$

$$\ddot{x} = -\frac{1}{100}(1000 + v^2)$$

(\beta) Let the max. height reached be H metres, $v = 0$

Using $v \frac{dv}{dx} = -\frac{1}{100}(1000 + v^2)$

$$\Rightarrow \frac{dv}{dx} = -\frac{(1000 + v^2)}{100v}$$

$$\frac{dx}{dv} = -\frac{100v}{1000 + v^2}$$

$$\Rightarrow dx = -\frac{100v}{1000 + v^2} dv$$

$$\int_0^H dx = -\int_5^0 \frac{100v}{1000 + v^2} dv$$

$$\therefore \int_0^H dx = \int_0^5 \frac{100v}{1000 + v^2} dv$$

$$\int_0^4 dx = \frac{1}{3} \int_0^5 \frac{300v}{1000+v^3} dv$$

$$= \frac{1}{3} \int_0^5 \left[\frac{-10}{v+10} + \frac{10v+100}{v^2-10v+100} \right] dv$$

from (i)

$$H = \frac{1}{3} \left[-10 \ln(v+10) \right]_0^5 + 10 \int_0^5 \frac{v+10}{v^2-10v+100} dv$$

$$= \frac{1}{3} \left[-10 \ln 15 + 10 \ln 10 \right] + 10 \int_0^5 \frac{\frac{1}{2}(2v-10)+15}{v^2-10v+100} dv$$

$$= \frac{1}{3} \left[10 \ln \frac{10}{15} + 5 \ln(v^2-10v+100) \right]_0^5 + \int_0^5 \frac{150}{75+(v-5)^2} dv$$

$$= \frac{1}{3} \left[10 \ln \frac{10}{15} + 5 \ln(75) - 5 \ln(100) + \frac{150}{5\sqrt{3}} \tan^{-1} \frac{(v-5)}{5\sqrt{3}} \right]_0^5$$

$$= \frac{1}{3} \left[10 \ln \frac{10}{15} + 5 \ln 75 - 5 \ln 100 + \frac{150}{5\sqrt{3}} \tan^{-1} 0 + \frac{150}{5\sqrt{3}} \left(\tan^{-1} \frac{5}{5\sqrt{3}} \right) \right]$$

$$= \frac{1}{3} \left[10 \ln \frac{2}{3} + 5 \ln 75 - 5 \ln 100 + \frac{150}{5\sqrt{3}} \times \frac{\pi}{6} \right]$$

$$= 1.19197 \dots$$

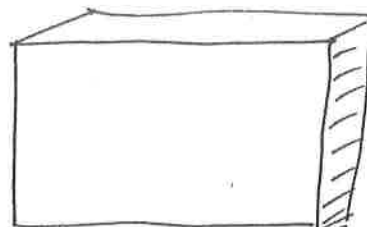
$$= 1.2 \text{ m (1 dp)}$$

b)

$$x = 4 - y^2 \Rightarrow y = \sqrt{4-x}$$

($y \geq 0$).

$$4-x = a$$



$$\frac{x}{4} = \frac{4-x}{4}$$

$$x = 4-x$$

$$\delta A = 2y(4-x) \delta x$$

$$\delta V = 2y(4-x) \delta x$$

Total Volume

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^4 2(4-x)y \delta x$$

$$= \lim_{\delta x \rightarrow 0} \sum_{x=0}^4 2(4-x)\sqrt{4-x} \delta x$$

$$= 2 \int_0^4 (4-x)^{3/2} dx$$

$$= 2 \left[\frac{2(4-x)^{5/2}}{-5} \right]_0^4$$

$$= -\frac{4}{5} (4-x)^2 \sqrt{4-x} \Big|_0^4$$

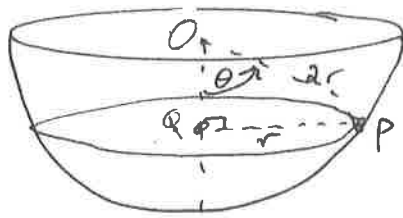
$$= \frac{4}{5} (4-x)^2 \sqrt{4-x} \Big|_4^0$$

$$= \frac{4}{5} (4)^2 \sqrt{4} - 0$$

$$= \frac{4 \times 16 \times 2}{5}$$

$$= \frac{128}{5} \text{ m}^3$$

16(1)

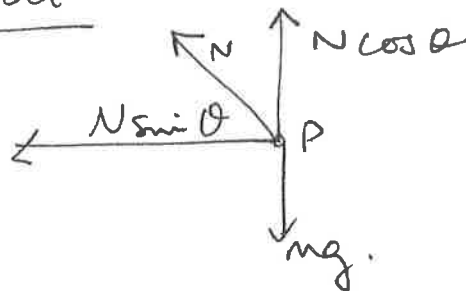


$$OQ^2 = (2r)^2 - r^2$$

$$= 3r^2$$

$$\therefore OQ = r\sqrt{3}$$

Vertically at P.



by Newton's Third Law $N \cos \theta - mg = 0$

$$\Rightarrow N \cos \theta = mg \quad (1) \quad \checkmark$$

Radially: $m\vec{r} = \sum \vec{F}_n$

$$m r \omega^2 = N \sin \theta \quad (2) \quad \checkmark$$

but in ΔOQP , $\frac{r}{2r} = \sin \theta$

$$\sin \theta = \frac{1}{2}$$

$$m r \omega^2 = \frac{N}{2}$$

$$N = 2m r \omega^2 \quad (3)$$

hence sub (3) in (1)

$$\Rightarrow 2m r \omega^2 \cos \theta = mg \quad \text{and } \cos \theta = \frac{r\sqrt{3}}{2r} = \frac{\sqrt{3}}{2} \quad \checkmark$$

$$\therefore 2m r \omega^2 \frac{\sqrt{3}}{2} = mg$$

$$r \omega^2 \sqrt{3} = g$$

$$\omega^2 = \frac{g}{r\sqrt{3}} \quad \text{as required.}$$